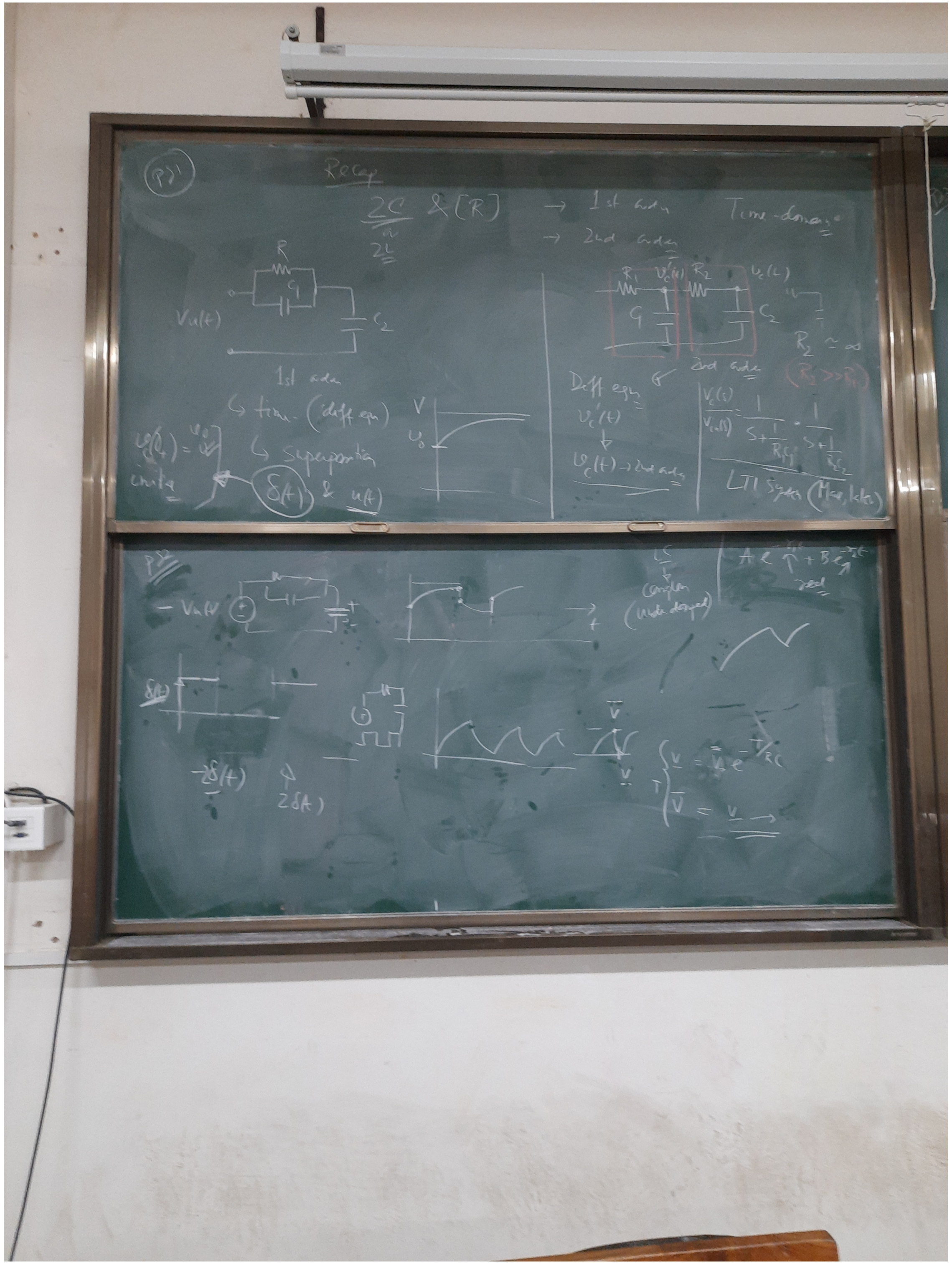


P1

Recap

$\underline{2C}$ & [R] → 1st order

or → 2nd order

$2L$

Time-domain

R
C_1
$V_{in}(t)$
C_2

1st order

↳ time (diff eqn)

↳ superposition

$v_c(0_+) = v_0$

initial

$\delta(t)$ & $u(t)$

V
v_0

R_1
$v_c'(t)$
R_2
$v_c(t)$
C_1
C_2

2nd order

$R_2 \simeq \infty$

$(R_2 >> R_1)$

Diff eqn

$v_c'(t)$

↓

$v_c(t) \to$ 2nd order

$$\frac{V_c(s)}{V_{in}(s)} = \frac{1}{s+\frac{1}{R_1C_1}} \cdot \frac{1}{s+\frac{1}{R_2C_2}}$$

LTI System (Linear, [illegible])

P2

$V_{in}(t)$
+
−
t

LC

↓

complex

(under damped)

$Ae^{-r_1t} + Be^{-r_2t}$

real

$\delta(t)$

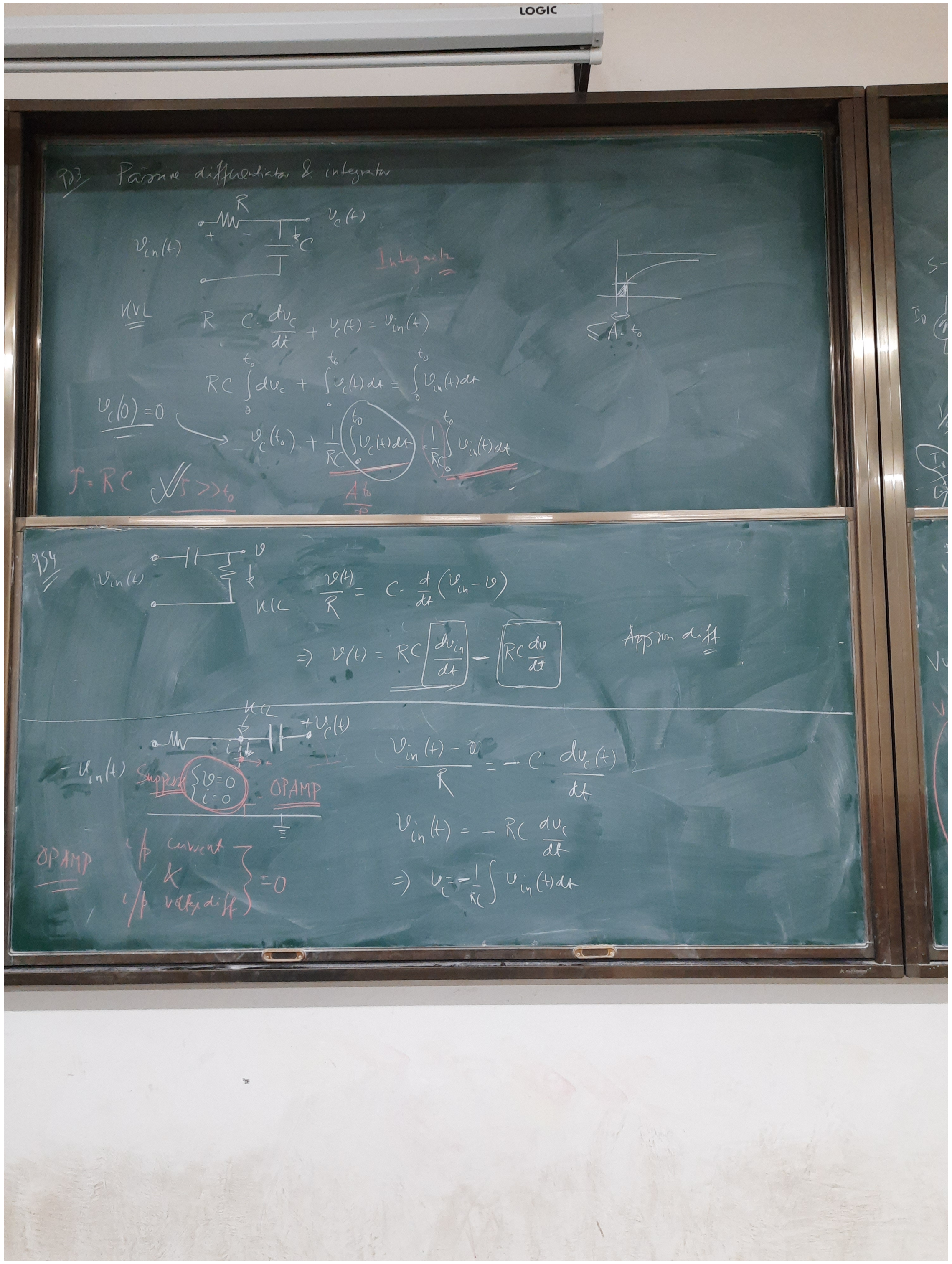
$-2\delta(t)$

$2\delta(t)$

$\overline{V}$
$\underline{V}$

$\underline{V} = \overline{V}e^{-T/RC}$

$\overline{V} = \underline{V} \rightarrow$



Q3 Passive differentiator & integrator

Integrator

KVL $R\ C\cdot\frac{dv_c}{dt} + v_c(t) = v_{in}(t)$

$RC\int_0^{t_0} dv_c + \int_0^{t_0} v_c(t)\,dt = \int_0^{t_0} v_{in}(t)\,dt$

$v_c(0)=0$ $\rightarrow$ $v_c(t_0) + \frac{1}{RC}\int_0^{t_0} v_c(t)\,dt = \frac{1}{RC}\int_0^{t_0} v_{in}(t)\,dt$

$\frac{At_0}{\tau}$

$\tau = RC$ $\tau \gg t_0$

Q4

KCL $\frac{v(t)}{R} = C\cdot\frac{d}{dt}(v_{in}-v)$

$\Rightarrow v(t) = RC\boxed{\frac{dv_{in}}{dt}} - \boxed{RC\frac{dv}{dt}}$

Approx diff

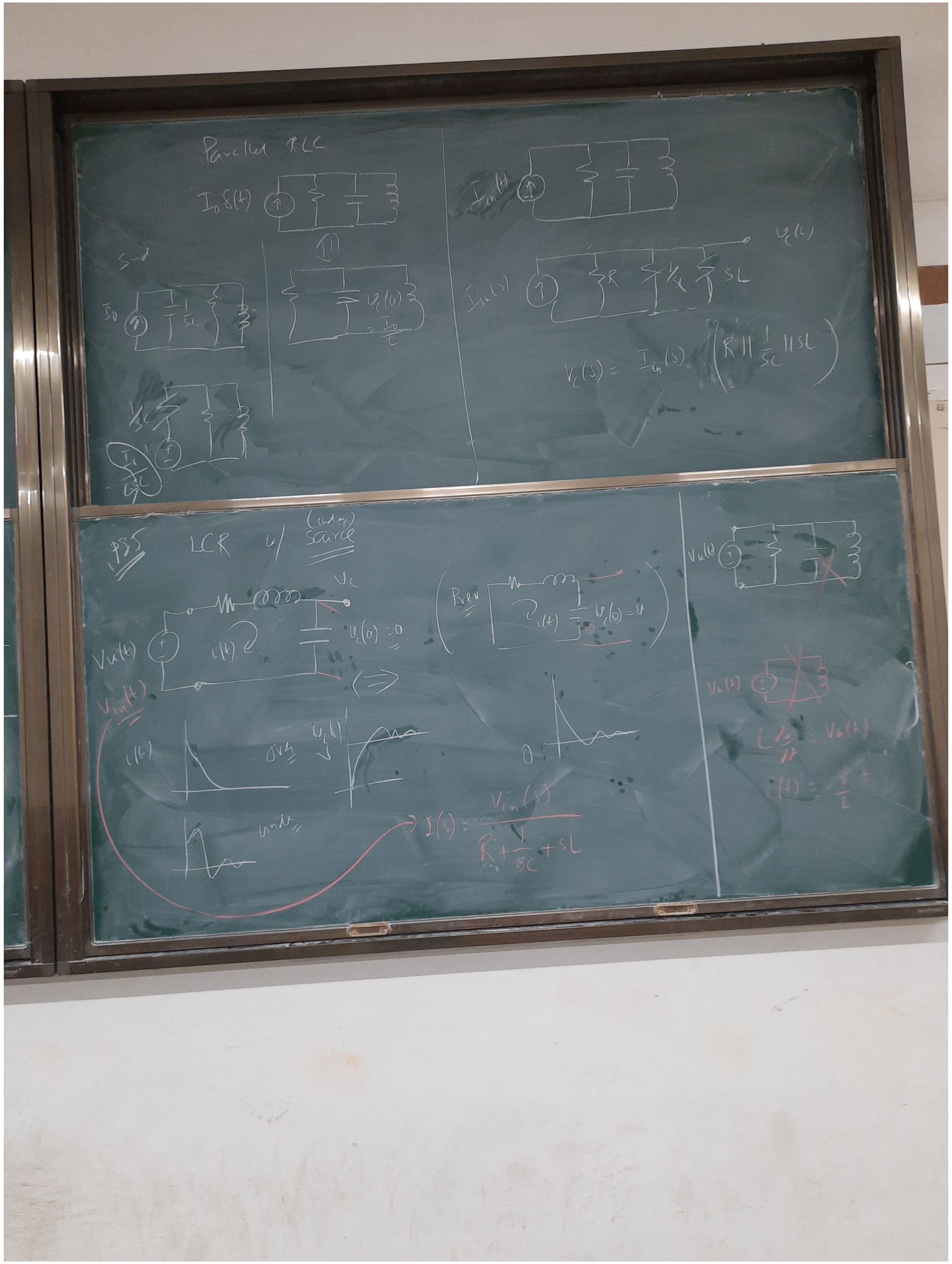
Suppose $\begin{cases} v=0 \\ i=0 \end{cases}$ OPAMP

$\frac{v_{in}(t)-v}{R} = -C\frac{dv_c(t)}{dt}$

$v_{in}(t) = -RC\frac{dv_c}{dt}$

$\Rightarrow v_c = -\frac{1}{RC}\int v_{in}(t)\,dt$

OPAMP i/p current & i/p voltage diff $\} = 0$


Parallel RLC
$I_0\delta(t)$
s-d
I_0
$\frac{1}{sC}$
$V_C(0) = \frac{I_0}{C}$
$1/sC$
$\frac{I_0}{sC}$
$I_{in}(t)$
$v_C(t)$
$I_{in}(s)$
R
$1/sC$
sL
PS5
LCR w/ (unitstep) source
v_C
$v_{in}(t)$
$i(t)$
$v_C(0) = 0$
Prev
$i(t)$
$v_C(0) = V$
$\Longleftrightarrow$
$V_{in}(t)$
$i(t)$
over
$v_C(t)$
0
under
$v_{in}(t)$

$V_C(s) = I_{in}(s)\left(R \parallel \frac{1}{sC} \parallel sL\right)$

$I(s) = \frac{V_{in}(s)}{R + \frac{1}{sC} + sL}$

$L\frac{di}{dt} = V_{in}(t)$

$i(t) = \frac{V}{L}t$